BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MIRPURKHAS

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A (MCQ's)

Note: Q.1 Choose the correct answer from the following choices:

Qi. Which form of energy is sound?

(a) electrical
(b) mechanical
(c) thermal
(d) chemical

Qii. Which of the following quantity is not changed during refraction of light?

(a) its direction
(b) its speed
(c) its frequency
(d) its wavelength

Qiii. A converging mirror with a radius of 20 cm creates a real image 30 cm from the mirror. What is the object distance?

(a) -5.0 cm
(b) -7.5 cm
(c) -15 cm
(d) -20 cm

Qiv. An object is placed at the centre of curvature of a concave mirror. The image produced by the mirror is located

(a) out beyond the centre of curvature.
(b) at the centre of curvature
(c) between the centre of curvature and the focal point
(d) at the focal point

Qv. The presence of a magnetic field can be detected by a

(a) small mass
(b) stationary positive charge
(c) stationary negative charge
(d) magnetic compass

Qvi. If the current in a wire which is placed perpendicular to a magnetic field increases, the force on the wire

(a) increases
(b) decreases
(c) remains the same
(d) will be zero

Qvii. The process by which electrons are emitted by a hot metal surface is known as

(a) boiling
(b) evaporation
(c) conduction
(d) thermionic emission

Qviii. When you rub a plastic rod against your hair several times and put it near some bits of paper, the pieces of papers are attracted towards it. What does this observation indicate?

(a) the rod and the paper are oppositely charged
(b) the rod acquires a positive charge
(c) the rod and the paper have the same charges
(d) the rod acquires a negative charge

Qix. According to Coulomb's law, what happens to the attraction of two oppositely charged objects as their distance of separation increases?

(a) increases
(b) decreases
(c) remains unchanged
(d) cannot be determined

Qx. Electric potential and e.m.f

(a) are the same terms
(b) are the different terms
(c) have different units
(d) both (b) and (c)

Qxi. Which of the following characteristics of a wave is independent of the others?

(a) speed
(b) frequency
(c) amplitude
(d) wavelength

Qxii. The relation between v, f and λ of a wave is

(a) $vf = \lambda$
(b) $f\lambda = v$
(c) $v\lambda = f$
(d) $v = \lambda/$

Qxiii. Image formed by a camera is

(a) real, inverted, and diminished
(b) virtual, upright and diminished
(c) virtual, upright and magnified
(d) real, inverted and magnified

Qxiv. If a ray of light in glass is incident on an air surface at an angle greater than the critical angle, the ray will

(a) refract only
(b) reflect only
(c) partially refract and partially reflect
(d) diffract only

Qxv. A positive and a negative charge are initially 4 cm apart. When they are moved closer together so that they are now only 1 cm apart, the force between them is

(a) 4 times smaller than before
(b) 4 times larger than before
(c) 8 times larger than before
(d) 16 times larger than before

Section-B (Short Answer)

> Note: Answer any TEN of the following questions.

Q2. What is simple harmonic motion? What are the necessary conditions for a body to execute simple harmonic motion?

Q3. What is gold leaf electroscope?

Q4. Why or why not concave mirrors are suitable for makeup?

Q5. How can you define the term wave?

Q6. A pendulum of length 0.99 m is taken to the Moon by an astronaut. The period of the pendulum is 4.9 s. What is the value of g on the surface of the Moon?

Q7. Why ultrasound is useful in medical field?

Q8. Describe the following terms used in reflection:

(i) normal

(ii) angle of incidence

(iii) angle of reflection

Q9. How does the thickness of a lens affect its focal length?

Q10. Two points on an object are at different electric potentials. Does charge necessarily flow between them?

Q11. What is difference between data and information?

Q12. Write the Note:

(i) Simple Harmonic Motion

(ii) Noise Pollution

(iii) Fax Machine

Section-C (Descriptive Answer)

> Note: Answer any THREE of the following questions. marks.

Q13.

(a) An electrified rod attracts pieces of paper. After a while these pieces fly away! Why?

(b) The electric potential at a point in an electric field is 10 V. If a charge of +100 μC is 4 brought from infinity to this point. What would be the amount of work done on it?

Q14. Explain Ohm's law. What are its limitations?

Q15. What are the components of information technology? Clearly indicate the function of each component.

Q16.

(a) What is the difference between hardware and software? Name different soft wares?

(b) Ashes from a campfire deep in a cave show carbon-14 activity of only one-eighth the activity of fresh wood. How long ago was that campfire made?

Q17. Write the note any one of the following:

(i) Information and Communication Technology

(ii) Ripple Tank

(iii) House Safety Alarm